# Autoregressive Work

Michael Bezick

April 24, 2026

TODO 9/29/2025
ALLERTON - June 8th deadline.
Find right identifiability condition / metric on phi.
Experiments

1. check that overtime that alphas become eigenvectors of phi
2. experiment that does model selection to find right value of p

Want plots where actual errors are going to 0. Train current setup longer
Check convexity of BTL
Try to see if there are rank guarantees of final $\widehat{A}$.

$$\forall i, max_{\alpha_{i,1},...,\alpha_{i,T},\phi_{i,1},...,\phi i,p} \prod_{t=1}^{T} \prod_{(i,j)} \left( \frac{e^{\alpha_{i,t}}}{e^{\alpha_{i,t}} + e^{\alpha_{j,t}}} \right)^{Z_{i,j,t}} \left( \frac{e^{\alpha_{j,t}}}{e^{\alpha_{i,t}} + e^{\alpha_{j,t}}} \right)^{W_{i,j,t}} \tag{1}$$

where

$$Z_{i,j,t} = \begin{cases} 1, & \text{if i beats j at time t} \\ 0, & \text{otw} \end{cases}, W_{i,j,t} = \begin{cases} 1, & \text{if j beats i at time t} \\ 0, & \text{otw} \end{cases} \tag{2}$$

subject to: $\forall t \in \{p+1, ..., n\}, \alpha_{i,t} = \sum_{j=1}^{p} \phi_{i,j} \alpha_{i,t-j}$

$$\text{BTL-likelihood} = \forall i, max_{\alpha_{i,1},...,\alpha_{i,T}} \log \left( \prod_{t=1}^{T} \prod_{(i,j)} \left( \frac{e^{\alpha_{i,t}}}{e^{\alpha_{i,t}} + e^{\alpha_{j,t}}} \right)^{Z_{i,j,t}} \left( \frac{e^{\alpha_{j,t}}}{e^{\alpha_{i,t}} + e^{\alpha_{j,t}}} \right)^{W_{i,j,t}} \right) \tag{3}$$

$$\text{AR-error} = \forall i, min_{\phi i,1,...,\phi i,p} \sum_{t=p+1}^{T} (\alpha_{i,t} - \sum_{k=1}^{p} \phi_{i,k} \alpha_{i,t-1-(p-k)})^2 \tag{4}$$

$$\text{Final equation: } max_{\{\alpha_{i,t}\},\{\phi_{i,k}\}} \text{BTL-likelihood}(\alpha) - \lambda \cdot \text{AR-error}(\alpha, \phi) \tag{5}$$

## 1 Autoregressive Process

$n$ players

At time step $t$, for player $j$, we have a skill level $\alpha_{j,t} \in \mathbb{R}$. Update equations are

$$\alpha_{j,t+1} = \sum_{i=0}^{p-1} \phi_{j,i} \alpha_{j,t-i} + n_{j,t}$$

where $n_t$ are i.i.d. zero-mean Gaussian noise, and player-specific parameters $\{\phi_{j,i+1} \in \mathbb{R}\}$. Consider a stacked (column) vector $\alpha_t \in \mathbb{R}^n$ of all skills levels at time $t$. Then, letting $\phi_i = [\phi_{1,i} \cdots \phi_{n,i}]^T$, we have

$$\alpha_{t+1} = \sum_{i=0}^{p-1} \text{diag}(\phi_i) \alpha_{t-i} + n_t \, .$$

Suppose $\alpha_0$ has the property that $1^T\alpha_0 = 0$ (i.e., average of $\alpha_0$ is 0) to ensure identifiability, where $1 = [1 \cdots 1]^T$ denotes the all-ones vector of appropriate dimension.

$$1^T\alpha_{t+1} = \sum_{i=0}^{p-1} \phi_i^T \alpha_{t-i} + \underbrace{1^T n_t}_{\text{zero-mean}} \approx \sum_{i=0}^{p-1} \phi_i^T \alpha_{t-i} \overset{\text{want}}{=} 0$$

Notice that is $\phi_i = 1$ for all $i$, we can ensure the average-constraint, but this renders the estimation question moot.

So, consider instead the following more general model:

$$\alpha_{t+1} = \sum_{i=0}^{p-1} \Phi_i \alpha_{t-i} + n_t$$

where $\Phi_i \in \mathbb{R}^{n\times n}$. Again, suppose $\alpha_0$ has the property that $1^T\alpha_0 = 0$. Assume further that $\Phi_i^T 1 = 1$ for all $i$ (i.e., columns sum to 1). Then, by induction,

$$1^T\alpha_{t+1} = \sum_{i=0}^{p-1} 1^T\Phi_i\alpha_{t-i} + \underbrace{1^T n_t}_{\text{zero-mean}} \approx \sum_{i=0}^{p-1} 1^T\Phi_i\alpha_{t-i} = \sum_{i=0}^{p-1} 1^T\alpha_{t-i} = 0$$

# 2 Algorithm

- Predict skill parameters $\alpha$ for fixed AR parameters $\phi$ with $N_{\text{grad\_descent}}$ steps.

- Get prediction of $\phi$ with fixed $\alpha$ with closed-form sum-of-squares solution, subject to column constraint.

# 3 Sum of Squares w/ Constraint Optimization

Our new parameter set is $\{\Phi_0, ..., \Phi_{p-1}\}$

$$\min_{\{\Phi_i\}} f_0(\{\Phi_i\}) = \min_{\Phi} \frac{1}{2} \sum_t \left\| \boldsymbol{\alpha}_{t+1} - \sum_{i=0}^{p-1} \Phi_i \boldsymbol{\alpha}_{t-i} \right\|_2^2$$

$$\text{Subject to } \ \Phi_i^T \mathbf{1} = \mathbf{1}$$

$$L(\{\Phi_i\}, \{\boldsymbol{\nu}_i\}) = f_0(\{\Phi_i\}) + \sum_{i=0}^{p-1} \boldsymbol{\nu}_i^T (\Phi_i^T \mathbf{1}_n - \mathbf{1}_n)$$

Let $I_k$ denote the $k \times k$ identity matrix and $\mathbf{1}_k$ denote the $k$-dimensional all one's vector.

**Calculation:** First, we simplify:

$$\begin{aligned}
L(\{\Phi_i\}, \{\boldsymbol{\nu}_i\}) &= \frac{1}{2} \sum_t \left( \boldsymbol{\alpha}_{t+1} - \sum_{i=0}^{p-1} \Phi_i \boldsymbol{\alpha}_{t-i} \right)^T \left( \boldsymbol{\alpha}_{t+1} - \sum_{i=0}^{p-1} \Phi_i \boldsymbol{\alpha}_{t-i} \right) + \sum_{i=0}^{p-1} \mathbf{1}_n^T (\Phi_i - I_n) \boldsymbol{\nu}_i \\
&= \frac{1}{2} \sum_t \boldsymbol{\alpha}_{t+1}^T \boldsymbol{\alpha}_{t+1} - 2 \sum_{i=0}^{p-1} \boldsymbol{\alpha}_{t+1}^T \Phi_i \boldsymbol{\alpha}_{t-i} + \left( \sum_{i=0}^{p-1} \boldsymbol{\alpha}_{t-i}^T \Phi_i^T \right) \left( \sum_{i=0}^{p-1} \Phi_i \boldsymbol{\alpha}_{t-i} \right) + \sum_{i=0}^{p-1} \mathbf{1}_n^T (\Phi_i - I_n) \boldsymbol{\nu}_i \\
&= \frac{1}{2} \sum_t \left( \boldsymbol{\alpha}_{t+1}^T \boldsymbol{\alpha}_{t+1} - 2 \sum_{i=0}^{p-1} \boldsymbol{\alpha}_{t+1}^T \Phi_i \boldsymbol{\alpha}_{t-i} + \sum_{i=0}^{p-1} \sum_{j=0}^{p-1} \boldsymbol{\alpha}_{t-i}^T \Phi_i^T \Phi_j \boldsymbol{\alpha}_{t-j} \right) + \sum_{i=0}^{p-1} \mathbf{1}_n^T (\Phi_i - I_n) \boldsymbol{\nu}_i
\end{aligned}$$

Next, we differentiate and set equal to 0:

$$\nabla_{\boldsymbol{\nu}_i} L(\{\Phi_i\}, \{\boldsymbol{\nu}_i\}) = (\Phi_i - I_n)^T \mathbf{1}_n = 0 \ \ \Leftrightarrow \ \ \Phi_i^T \mathbf{1}_n = \mathbf{1}_n$$

$$\nabla_{\boldsymbol{\Phi}_i} L(\{\Phi_i\}, \{\boldsymbol{\nu}_i\}) = \sum_{k=0}^{p-1} \Phi_k \left( \sum_t \boldsymbol{\alpha}_{t-k} \boldsymbol{\alpha}_{t-i}^T \right) - \left( \sum_t \boldsymbol{\alpha}_{t+1} \boldsymbol{\alpha}_{t-i}^T \right) + \mathbf{1}_n \boldsymbol{\nu}_i^T = 0$$

which is equivalent to

$$\sum_{k=0}^{p-1} \Phi_k \underbrace{\left(\sum_t \boldsymbol{\alpha}_{t-k} \boldsymbol{\alpha}_{t-i}^T\right)}_{A_{k,i}} = \underbrace{\left(\sum_t \boldsymbol{\alpha}_{t+1} \boldsymbol{\alpha}_{t-i}^T\right)}_{A_{-1,i}} - \mathbf{1}_n \boldsymbol{\nu}_i^T$$

$$\underbrace{\begin{bmatrix}\Phi_0 & \Phi_1 & \cdots & \Phi_{p-1}\end{bmatrix}}_{B} \underbrace{\begin{bmatrix} A_{0,i} \\ A_{1,i} \\ \vdots \\ A_{p-1,i}\end{bmatrix}}_{a_i} = A_{-1,i} - \mathbf{1}_n \boldsymbol{\nu}_i^T$$

Then each $i$th equation is

$$Ba_i = A_{-1,i} - \mathbf{1}_n \boldsymbol{\nu}_i^\top.$$

Stack these equations horizontally over $i$. Define

$$\mathcal{A} \coloneqq \begin{bmatrix} A_{0,0} & A_{0,1} & \cdots & A_{0,p-1} \\ A_{1,0} & A_{1,1} & \cdots & A_{1,p-1} \\ \vdots & \vdots & \ddots & \vdots \\ A_{p-1,0} & A_{p-1,1} & \cdots & A_{p-1,p-1}, \end{bmatrix} \in \mathbb{R}^{pn \times pn}$$

$$\mathcal{R} \coloneqq \begin{bmatrix} A_{-1,0} & A_{-1,1} & \cdots & A_{-1,p-1}\end{bmatrix} \in \mathbb{R}^{n \times pn},$$

$$V \coloneqq \begin{bmatrix} \boldsymbol{\nu}_0^\top & \boldsymbol{\nu}_1^\top & \cdots \boldsymbol{\nu}_{p-1}^\top\end{bmatrix} \in \mathbb{R}^{1 \times pn}.$$

Stacking over $i$ gives

$$B\mathcal{A} = \mathcal{R} - \mathbf{1}_n V,$$
$$B = (\mathcal{R} - \mathbf{1}_n V)\mathcal{A}^{-1}.$$

Apply constraints

$$\Phi_i^\top \mathbf{1}_n = \mathbf{1}_n,$$

stacking as

$$B^\top \mathbf{1}_n = \begin{bmatrix} \Phi_0^\top \mathbf{1}_n \\ \Phi_1^\top \mathbf{1}_n \\ \vdots \\ \Phi_{p-1}^\top \mathbf{1}_n \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{1}_n \\ \mathbf{1}_n \\ \vdots \\ \mathbf{1}_n \end{bmatrix}}_{\mathbf{1}_p \otimes \mathbf{1}_n \in \mathbb{R}^{pn}}.$$

Substitute $B^\top$,

$$\begin{aligned}
\left((\mathcal{R} - \mathbf{1}_n V)\mathcal{A}^{-1}\right)^\top \mathbf{1}_n &= \mathbf{1}_p \otimes \mathbf{1}_n \\
(\mathcal{A}^{-1})^\top (\mathcal{R}^\top - V^\top \mathbf{1}_n^\top)\mathbf{1}_n &= \mathbf{1}_p \otimes \mathbf{1}_n \\
(\mathcal{A}^{-1})^\top (\mathcal{R}^\top \mathbf{1}_n - nV^\top) &= \mathbf{1}_p \otimes \mathbf{1}_n & \mathbf{1}_n^\top \mathbf{1}_n = n \\
A^\top (\mathcal{A}^{-1})^\top (\mathcal{R}^\top \mathbf{1}_n - nV^\top) &= \mathcal{A}^\top \mathbf{1}_p \otimes \mathbf{1}_n \\
(\mathcal{A}^{-1} A)^\top (\mathcal{R}^\top \mathbf{1}_n - nV^\top) &= \mathcal{A}^\top \mathbf{1}_p \otimes \mathbf{1}_n \\
\mathcal{R}^\top \mathbf{1}_n - nV^\top &= \mathcal{A}^\top \mathbf{1}_p \otimes \mathbf{1}_n \\
V^\top &= \frac{1}{n}(\mathcal{R}^\top \mathbf{1}_n - \mathcal{A}^\top (\mathbf{1}_p \otimes \mathbf{1}_n)) \\
V &= \frac{1}{n}(\mathbf{1}_n^\top \mathcal{R} - (\mathbf{1}_p \otimes \mathbf{1}_n)^\top \mathcal{A}).
\end{aligned}$$

Plugging back into the original equation,

$$B = \left(\mathcal{R} - \mathbf{1}_n \frac{1}{n}(\mathbf{1}_n^\top \mathcal{R} - (\mathbf{1}_p \otimes \mathbf{1}_n)^\top \mathcal{A})\right)\mathcal{A}^{-1}$$
$$B = \mathcal{R}\mathcal{A}^{-1} - \frac{1}{n}\mathbf{1}_n \mathbf{1}_n^\top \mathcal{R}\mathcal{A}^{-1} + \frac{1}{n}\mathbf{1}_n (\mathbf{1}_p \otimes \mathbf{1}_n)^\top$$

After doing this, I get a successful experiment:

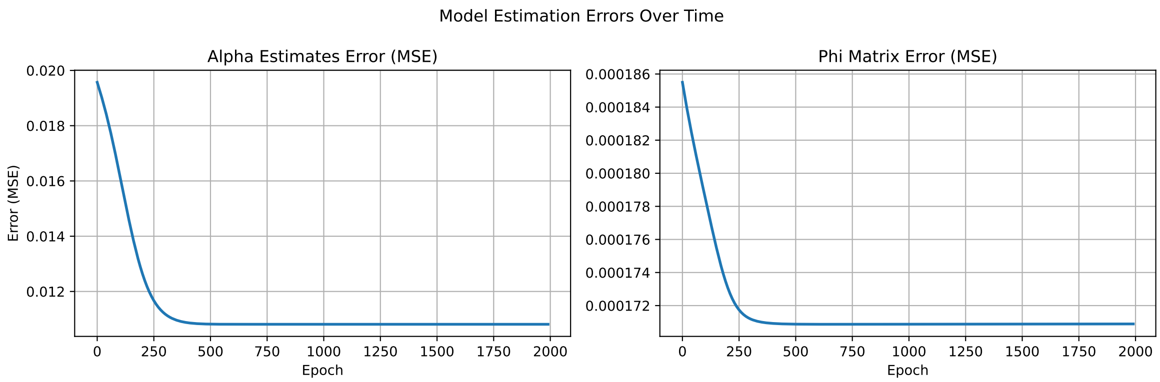



Figure 1: Least squares with constraints

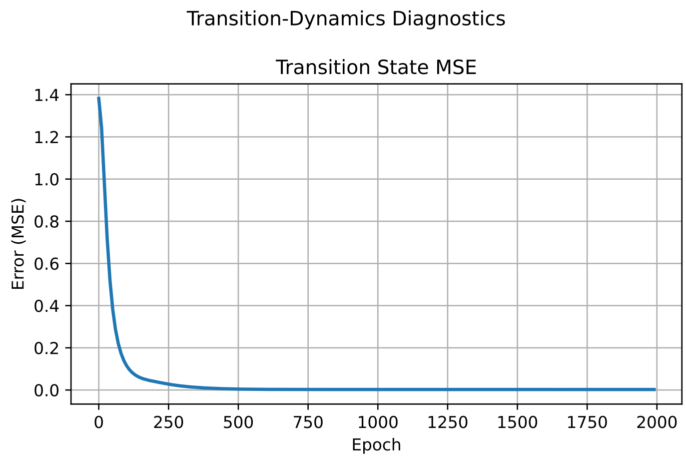



Figure 2: Experiment with new identifiability metric. Plug in true latent skill to model, then calculate MSE with true next latent skill, with identity matrix AR model.

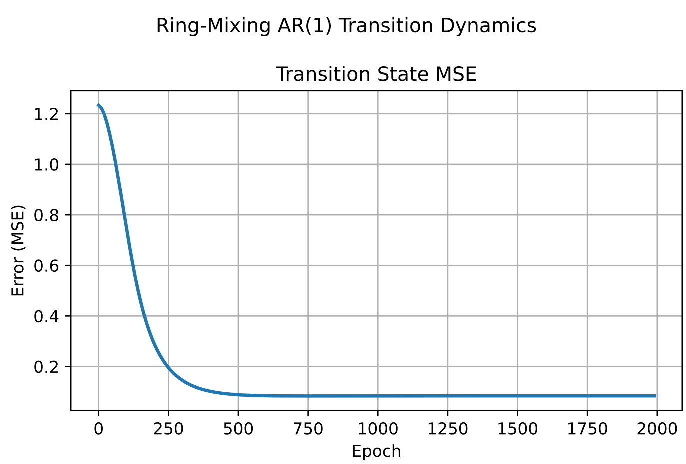



Figure 3: Experiment with new identifiability metric. Plug in true latent skill to model, then calculate MSE with true next latent skill, with ring shift AR model.

# 4 Model Selection Experiment

75-25 split, model order p selected based off best predictive performance on held out dataset.

```
true_p=1: hist={"8": 4, "10": 6}
true_p=2: hist={"1": 1, "5": 1, "8": 2, "10": 6}
true_p=4: hist={"6": 1, "8": 2, "10": 7}
true_p=8: hist={"1": 1, "8": 3, "10": 6}
```

## 5 Check if Likelihood function is concave

Assuming all pairwise games for $j > i$ are observed, with 1 game per pair:

$$\log \mathcal{L}_{\text{BTL}} + \lambda \log \mathcal{L}_{\text{AR}} = \sum_{(i,j)} Z_{i,j,t} \log(\frac{e^{\alpha_{i,t}}}{e^{\alpha_{i,t}} + e^{\alpha_{j,t}}}) + W_{i,j,t} \log(\frac{e^{\alpha_{j,t}}}{e^{\alpha_{i,t}} + e^{\alpha_{j,t}}}) - \lambda \sum_{t=p+1}^{T} \left\| \boldsymbol{\alpha}_t - \sum_{k=1}^{p} \Phi_k \boldsymbol{\alpha}_{t-k} \right\|_2^2 \tag{6}$$

Define residual $\boldsymbol{r}_t = \boldsymbol{\alpha}_t - \sum_{k=1}^{p} \Phi_k \boldsymbol{\alpha}_{t-k}$. Stack all skill parameters and residuals

$$\boldsymbol{\alpha} = \begin{bmatrix} \boldsymbol{\alpha}_1 \\ \vdots \\ \boldsymbol{\alpha}_T \end{bmatrix}, \quad \boldsymbol{r} = \begin{bmatrix} \boldsymbol{r}_{p+1} \\ \vdots \\ \boldsymbol{r}_T \end{bmatrix}.$$

Because each residual $\boldsymbol{r}_t$ is linear in $\alpha$, there exists a matrix G such that

$$\boldsymbol{r} = G\boldsymbol{\alpha}.$$

Then, the AR likelihood term becomes

$$\begin{aligned}
\log \mathcal{L}_{AR} &= - \sum_{t=p+1}^{T} \|\boldsymbol{r}_t\|_2^2 \\
&= - \|\boldsymbol{r}\|_2^2 \\
&= - \|G\boldsymbol{\alpha}\|_2^2 \\
&= -\boldsymbol{\alpha}^\top G^\top G \boldsymbol{\alpha} \\
\nabla_{\boldsymbol{\alpha}} \log \mathcal{L}_{AR} &= -2G^\top G \boldsymbol{\alpha} \\
\nabla^2_{\boldsymbol{\alpha}} \log \mathcal{L}_{AR} &= -2G^\top G,
\end{aligned}$$

which is negative semi-definite.

## 6 Things to Prove

Prove some subset of these results.

1. Algorithmic Convergence: As iterates tend to infinity, the algorithm converges to some solutions (may not be optimal). [Proof idea: Show that the GD and least-squares steps always decrease the objective function. This means the objective function values form a monotone sequence, which must converge in the extended reals.]

   *Proof.* We want to minimize - $\mathcal{L}_{BTL} + \mathcal{L}_{AR}$ using the alternating updating approach: (i) Using GD to update $\alpha$ (ii) Using the least square solution to update $\Phi$. Since the close-form update will never increase the loss, it suffices to show that the loss is monotonic decreasing during gradient decent. Let $f(\alpha) = -\mathcal{L}_{BTL} + \mathcal{L}_{AR}$. Assume that $\nabla_\alpha f$ is $L$-Lipchitz, and $L$ will be bounded later. For $\alpha, \alpha'$, we have

   $$f(\alpha') \leq f(\alpha) + \langle \nabla_\alpha f, \alpha' - \alpha \rangle + \frac{L}{2}||\alpha' - \alpha||_2 \qquad \text{Here } \alpha \text{ represents vector-flattened parameters} \tag{7}$$

   In gradient descent, we choose $\alpha' = \alpha - \eta\ \nabla_\alpha f(\alpha)$. Plug in and we get

   $$f(\alpha') = f(\alpha - \eta\ \nabla_\alpha f(\alpha)) \leq f(\alpha) - \eta(1 - \frac{L\eta}{2})||\nabla_\alpha f(\alpha)||_2^2 \tag{8}$$

   If we impose the learning rate $\eta \in (0, \frac{2}{L})$, $f(\alpha)$ will be monotonic decreasing in GD. The remaining thing is to upper bound the $L$. We will bound $L$ for BTL part and AR part separately.

   For BTL, consider the hessian with timestamp $t$:

   $$H_t = \sum_{(i,j)\in E_t} \sigma(\cdot)(1-\sigma(\cdot))(e_i - e_j)(e_i - e_j)^\top \preceq \frac{1}{4}L_{G_t} \quad L_{G_t} \text{ is the graph Laplacian for data at time t} \tag{9}$$

   So we have $\forall t, ||H_t||_2 \leq \frac{1}{4}||L_{G_t}||_2 \leq \frac{1}{2}d_{max,t}$, where $d_{max,t}$ denotes the largest degree in $L_{G_t}$. Stacking all $t$ together, we get $L_{BTL} = ||H||_2 \leq \frac{1}{2}\max_t d_{max,t}$.

   For AR part, $\mathcal{L}_{AR}(\alpha) = \sum_{t=p}^{T-1} \left\| \alpha_{t+1} - \sum_{k=0}^{p-1} \Phi_k \alpha_{t-k} \right\|_2^2, \alpha \in \mathbb{R}^{nT}$. Stacking all the residuals together, we get $r = \begin{bmatrix} r_p \\ r_{p+1} \\ \vdots \\ r_{T-1} \end{bmatrix} \in \mathbb{R}^{n(T-p)}$. Since each entry in $r$ is linear with $\alpha$, there exists a matrix $A_\Phi \in \mathbb{R}^{n(T-p)\times nT}$ that depends on $\Phi$, such that $r = A_\Phi \alpha$ and $\mathcal{L}_{AR}(\alpha) = ||A_\Phi \alpha||_2^2$. The hessian is:

   $$\nabla^2 \mathcal{L}_{AR}(\alpha) = 2A_\Phi^\top A_\Phi \qquad \text{and} \qquad L_{AR} = 2||A_\Phi||_2^2 \tag{10}$$

   Here the task is to upper bound $||A_\Phi||_2$, notice that $\forall t$:

   $$||r_t||_2 \leq ||\alpha_{t+1}||_2 + \sum_{k=0}^{p-1} ||\Phi_k||_2 ||\alpha_{t-k}||_2 \tag{11}$$

   By Cauchy-Schwarz,

   $$||r_t||_2^2 \leq (||\alpha_{t+1}||_2 + \sum_{k=0}^{p-1} ||\Phi_k||_2 ||\alpha_{t-k}||_2)^2 \leq (1+p)(||\alpha_{t+1}||_2^2 + \sum_{k=0}^{p-1} ||\Phi_k||_2^2 ||\alpha_{t-k}||_2^2) \tag{12}$$

Therefore,

$$
\begin{aligned}
||A_\Phi \alpha||_2^2 &= ||r||_F && (13)\\
&= \sum_{t=p}^{T-1} ||r_t||_2^2 && (14)\\
&\leq (1+p)\sum_{t=p}^{T-1}(||\alpha_{t+1}||_2^2 + \sum_{k=0}^{p-1}||\Phi_k||_2^2||\alpha_{t-k}||_2^2) && (15)\\
&= (1+p)(\sum_{t=p}^{T-1}||\alpha_{t+1}||_2^2 + \sum_{k=0}^{p-1}||\Phi_k||_2^2\sum_{t=p}^{T-1}||\alpha_{t-k}||_2^2) && (16)\\
&\leq (1+p)||\alpha||_2^2(1+\sum_{k=0}^{p-1}||\Phi_k||_2^2) && (17)
\end{aligned}
$$

Hence we get $||A_\Phi||_2^2 \leq (1+p)(1+\sum_{k=0}^{p-1}||\Phi_k||_2^2)$ and $L_{AR} \leq 2(1+p)(1+\sum_{k=0}^{p-1}||\Phi_k||_2^2)$. Combining the two parts, we get

$$
L \leq \frac{1}{2}\max_t d_{max,t} + 2\lambda(1+p)(1+\sum_{k=0}^{p-1}||\Phi_k||_2^2) := L_0 \tag{18}
$$

which yields for $\eta \in (0, \frac{2}{L_0})$, the loss is monotonic decreasing. $f(\alpha)$ is lower bounded by 0 which completes the proof. □

2. Algorithmic Complexity: Prove some bound on speed of convergence, e.g., after $S$ steps, the algorithm is within distance $\epsilon$ of the convergent point.

   **Remarks** The target function is probably not global jointly convex on $\alpha$ and $\Phi$. Now what we can get is for $\eta \in (0, \frac{1}{L_0})$, $f(\alpha^m, \Phi^m) - f(\alpha^{m+1}, \Phi^{m+1}) \geq \frac{\eta}{2}||\nabla_\alpha f(\alpha^m, \Phi^m)|||_2^2$. Given step $S$, we have $\sum_{m=0}^{S-1}||\nabla_\alpha f(\alpha^m, \Phi^m)|||_2^2 \leq \frac{2(f(\alpha^0,\Phi^0)-f_{inf})}{\eta}$. That is, for sufficiently large $S = O(\frac{1}{\eta\ \epsilon^2})$, we are guaranteed to get some $m$, such that $||\nabla_\alpha f(\alpha^m, \Phi^m)||_2 \leq \epsilon$. To make further analysis, we may need some kind of local strong convexity property around stationary point.

3. Statistical Consistency: The convergent point is the true parameters in the limit, e.g., on expectation or high probability. (Also pay attention to the identifiablility)

   **Fix $\Phi$, evaluation on $\alpha$** For fixed $\Phi$, if we assume for some $b \in (0,1)$, $\alpha_t \in [-b, b]^n, \forall t$ and then $-\frac{d^2}{dt^2}\sigma(t) \geq a > 0$ for $t \in [-2b, 2b]$. If we also have $\min_t \lambda_2(L_t) > 0$, the negative log-likelihood of BTL part is $\mu_\alpha$-strong convex on $\alpha$, $\mu_\alpha = ca\min_t \lambda_2(L_t) > 0$, and the AR part is convex on $\alpha$. So the whole function is $\mu_\alpha$-strong convex on $\alpha$.

$$
(\nabla_\alpha \mathcal{L}(\alpha, \Phi) - \nabla_\alpha \mathcal{L}(\alpha', \Phi))^\top(\alpha - \alpha') \geq \mu_\alpha ||\alpha - \alpha'||_2^2 \tag{19}
$$

   We need Lipchitz on $\phi$.

   **Lemma 6.1.** *There exists some $C > 0$:*

$$
||\nabla_\alpha \mathcal{L}(\alpha, \Phi) - \nabla_\alpha \mathcal{L}(\alpha, \Phi^*)||_2 \leq C||\Phi - \Phi^*||_{op} \tag{20}
$$

   *Proof.*

$$
\begin{aligned}
||\nabla_\alpha \mathcal{L}(\alpha, \Phi) - \nabla_\alpha \mathcal{L}(\alpha, \Phi^*)||_2 &= \lambda||\nabla_\alpha \mathcal{L}_{AR}(\alpha, \Phi) - \nabla_\alpha \mathcal{L}_{AR}(\alpha, \Phi^*)||_2 && (21)\\
&= 2\lambda||(A_\Phi^\top A_\Phi - A_{\Phi^*}^\top A_{\Phi^*})\alpha||_2 && (22)\\
&= 2\lambda|(A_\Phi^\top(A_\Phi - A_{\Phi^*}) - (A_{\Phi^*} - A_\Phi)^\top A_{\Phi^*}||_{op}||\alpha||_2 && (23)\\
&\leq 2\lambda||A_\Phi - A_{\Phi^*}||_{op}(||A_\Phi||_{op} + ||A_{\Phi^*}||_{op})||\alpha||_2 && (24)
\end{aligned}
$$

Since $A$ is linear on $\Phi$, we have

$$||(A_\Phi - A_{\Phi^*})\alpha||_2^2 = \sum_{t=p}^{T-1} || \sum_{k=0}^{p-1} (\Phi_k - \Phi_k^*)\alpha_{t-k}||_2^2 \tag{25}$$

$$\leq p \sum_{t=p}^{T-1} \sum_{k=0}^{p-1} ||\Phi_k - \Phi_k^*||_{op}^2 ||\alpha_{t-k}||_2^2 \qquad \text{By Cauchy-Schwarz} \tag{26}$$

$$= p \sum_{k=0}^{p-1} ||\Phi_k - \Phi_k^*||_{op}^2 \sum_{t=p}^{T-1} ||\alpha_{t-k}||_2^2 \tag{27}$$

$$\leq p \sum_{k=0}^{p-1} ||\Phi_k - \Phi_k^*||_{op}^2 ||\alpha||_2^2 \tag{28}$$

$$\leq p^2 ||\Phi - \Phi^*||_{op}^2 ||\alpha||_2^2 \tag{29}$$

This yields $||A_\Phi - A_{\Phi^*}||_{op} \leq p||\Phi - \Phi^*||_{op}$. If we assume $||A_\Phi||_{op}$ and $||\alpha_t||_2$ is bounded by some constant $B_\Phi \geq 0, \forall \Phi$ and $B_\alpha, \forall t$, choosing $C = 4\lambda p B_\Phi B_\alpha T$ completes the proof.

TODO: Use $||A_\Phi||_{op} \leq ||A_{\Phi^*}||_{op} + ||A_\Phi - A_{\Phi^*}||_{op} \leq ||A_{\Phi^*}||_{op} + p||\Phi - \Phi^*||_{op}$, only assume boundedness of $||A_{\Phi^*}||_{op}$

TODO: To get boundedness of $\alpha_t$'s in $\ell^\infty$ or $\ell^2$-norm, modify both dynamics of our model and the GD algorithm to use projection to the ball. □

Assume GD is run for infinitely many steps.

From the convexity and Lipchitzness, $\mu_\alpha ||\alpha - \alpha^*||_2^2 \leq (\nabla_\alpha \mathcal{L}(\alpha, \Phi) - \nabla_\alpha \mathcal{L}(\alpha^*, \Phi))^\top (\alpha - \alpha^*) \leq ||\nabla_\alpha \mathcal{L}(\alpha^*, \Phi))^\top||_2 ||\alpha - \alpha^*||_2$. Moreover,

$$\mu_\alpha ||\alpha - \alpha^*||_2 \leq ||\nabla_\alpha \mathcal{L}(\alpha^*, \Phi)^\top||_2 \tag{30}$$

$$\leq ||\nabla_\alpha \mathcal{L}(\alpha^*, \Phi)^\top - \nabla_\alpha \mathcal{L}(\alpha^*, \Phi^*)^\top||_2 + ||\nabla_\alpha \mathcal{L}(\alpha^*, \Phi^*)^\top||_2 \tag{31}$$

$$\leq 4\lambda p B_\Phi B_\alpha T ||\Phi - \Phi^*||_{op} + ||\nabla_\alpha \mathcal{L}(\alpha^*, \Phi^*)^\top||_2 \tag{32}$$

which infers a bound on statistical error on $\alpha$ basing the error on $\Phi$,

$$||\alpha - \alpha^*||_2 \leq \frac{4\lambda p B_\Phi B_\alpha T}{\mu_\alpha} ||\Phi - \Phi^*||_{op} + \frac{||\nabla_\alpha \mathcal{L}(\alpha^*, \Phi^*)^\top||_2}{\mu_\alpha} \tag{33}$$

**Fix $\alpha$, evaluation on $\Phi$** For fix $\alpha$, $\Phi$ can be directly calculated according to previous analysis

$$\Phi(\alpha)^\top = \mathcal{R}(\alpha)\mathcal{A}(\alpha)^{-1} - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top \mathcal{R}(\alpha)\mathcal{A}(\alpha)^{-1} + \frac{1}{n}\mathbf{1}_n(\mathbf{1}_p \otimes \mathbf{1}_n)^\top \tag{34}$$

We first derive bound on $||\Phi(\alpha) - \Phi(\alpha^*)||_{op}$

$$||\Phi(\alpha) - \Phi(\alpha^*)||_{op} = ||\Phi(\alpha)^\top - \Phi(\alpha^*)^\top||_{op} \tag{35}$$

$$= ||(I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top)(\mathcal{R}(\alpha)\mathcal{A}(\alpha)^{-1} - \mathcal{R}(\alpha^*)\mathcal{A}(\alpha^*)^{-1})||_{op} \tag{36}$$

$$\leq ||\mathcal{R}(\alpha)\mathcal{A}(\alpha)^{-1} - \mathcal{R}(\alpha^*)\mathcal{A}(\alpha^*)^{-1}||_{op} \tag{37}$$

$$= ||(\mathcal{R}(\alpha) - \mathcal{R}(\alpha^*))\mathcal{A}(\alpha)^{-1} + \mathcal{R}(\alpha^*)(\mathcal{A}(\alpha)^{-1} - \mathcal{A}(\alpha^*)^{-1})||_{op} \tag{38}$$

$$\leq ||\mathcal{R}(\alpha) - \mathcal{R}(\alpha^*)||_{op}||\mathcal{A}(\alpha)^{-1}||_{op} + ||\mathcal{R}(\alpha^*)||_{op}||\mathcal{A}(\alpha)^{-1} - \mathcal{A}(\alpha^*)^{-1}||_{op} \tag{39}$$

Assume $\lambda_{min}(\mathcal{A}(\alpha)) \geq \lambda_\mathcal{A} > 0$ Think about the invertibility - we need invertibility for all $\alpha$, but lower bound on eigenvalue for $\alpha^*$ - try to use bound in arXiv paper. Since $\mathcal{A}(\alpha)$ is symmetric, $||\mathcal{A}(\alpha)^{-1}||_{op} \leq \frac{1}{\lambda_\mathcal{A}}$.

$$||\mathcal{A}(\alpha)^{-1} - \mathcal{A}(\alpha^*)^{-1}||_{op} = ||(\mathcal{A}(\alpha)^{-1}\mathcal{A}(\alpha^*) - \mathcal{A}(\alpha)^{-1}\mathcal{A}(\alpha))\mathcal{A}(\alpha^*)^{-1}||_{op} \tag{40}$$

$$= ||\mathcal{A}(\alpha)^{-1}(\mathcal{A}(\alpha^*) - \mathcal{A}(\alpha))\mathcal{A}(\alpha^*)^{-1}||_{op} \tag{41}$$

$$\leq ||\mathcal{A}(\alpha)^{-1}||_{op}||\mathcal{A}(\alpha) - \mathcal{A}(\alpha^*)||_{op}||\mathcal{A}(\alpha^*)^{-1}||_{op} \tag{42}$$

$$\leq \frac{1}{\lambda_\mathcal{A}^2}||\mathcal{A}(\alpha) - \mathcal{A}(\alpha^*)||_{op} \tag{43}$$

Assume $||\alpha_t||_2$ Notice that

$$||\mathcal{A}(\alpha)_{k,i} - \mathcal{A}(\alpha^*)_{k,i}||_{op} = ||\sum_{t=0}^{T-1}(\alpha_{t-k}\alpha_{t-i}^\top - \alpha^*_{t-k}\alpha^{*\top}_{t-i})||_{op} \tag{44}$$

$$\leq \sum_{t=0}^{T-1}(||\alpha_{t-k} - \alpha^*_{t-k}||_2||\alpha_{t-i}||_2 + ||\alpha^*_{t-k}||_2||\alpha_{t-i} - \alpha^*_{t-i}||_2) \tag{45}$$

$$\leq B_\alpha \sum_{t=0}^{T-1}(||\alpha_{t-k} - \alpha^*_{t-k}||_2 + ||\alpha_{t-i} - \alpha^*_{t-i}||_2) \tag{46}$$

$$\leq 2B_\alpha\sqrt{T}||\alpha - \alpha^*||_2 \qquad \text{By Cauchy-Schwarz} \tag{47}$$

Hence, $||\mathcal{A}(\alpha) - \mathcal{A}(\alpha^*)||_{op} \leq 2B_\alpha p\sqrt{T}||\alpha - \alpha^*||_2$. Similarly,

$$||\mathcal{R}(\alpha)_i - \mathcal{R}(\alpha^*)_i||_{op} = ||\mathcal{A}(\alpha)_{-1,i} - \mathcal{A}(\alpha^*)_{-1,i}||_{op} \leq 2B_\alpha\sqrt{T}||\alpha - \alpha^*||_2 \tag{48}$$

which yields

$$||\mathcal{R}(\alpha) - \mathcal{R}(\alpha^*)||_{op} \leq (\sum_{i=0}^{p-1}||\mathcal{R}(\alpha)_i - \mathcal{R}(\alpha^*)_i||^2_{op})^{\frac{1}{2}} \leq 2B_\alpha\sqrt{pT}||\alpha - \alpha^*||_2 \tag{49}$$

$$||\mathcal{R}(\alpha^*)||_{op} \leq (\sum_{i=0}^{p-1}||\mathcal{A}(\alpha^*)_{-1,i}||^2_{op})^{\frac{1}{2}} \leq B_\alpha^2 T\sqrt{p} \tag{50}$$

Combine these inequalities together, we have

$$||\Phi(\alpha) - \Phi(\alpha^*)||_{op} \leq ||\mathcal{R}(\alpha) - \mathcal{R}(\alpha^*)||_{op}||\mathcal{A}(\alpha)^{-1}||_{op} + ||\mathcal{R}(\alpha^*)||_{op}||\mathcal{A}(\alpha)^{-1} - \mathcal{A}(\alpha^*)^{-1}||_{op} \tag{51}$$

$$\leq \frac{2B_\alpha\sqrt{pT}}{\lambda_\mathcal{A}}||\alpha - \alpha^*||_2 + \frac{2B_\alpha^3\sqrt{p^3T^3}}{\lambda_\mathcal{A}^2}||\alpha - \alpha^*||_2 \tag{52}$$

Then we need to bound $||\Phi(\alpha^*) - \Phi^*||_{op}$. Since $\Phi^{*T}\mathbf{1}_n = \mathbf{1}_p \otimes \mathbf{1}_n$, we can write $\Phi^* = (I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top)\Phi^* + \frac{1}{n}\mathbf{1}_n(\mathbf{1}_p \otimes \mathbf{1}_n)^\top$.

$$||\Phi(\alpha^*) - \Phi^*||_{op} = ||(I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top)(R(\alpha^*)\mathcal{A}(\alpha^*)^{-1} - \Phi^*)||_{op} \tag{53}$$

$$\leq ||R(\alpha^*) - \Phi^*\mathcal{A}(\alpha^*)||_{op}||\mathcal{A}(\alpha^*)^{-1}||_{op} \tag{54}$$

$$\leq \frac{1}{\lambda_\mathcal{A}}||R(\alpha^*) - \Phi^*\mathcal{A}(\alpha^*)||_{op} \tag{55}$$

where $R(\alpha^*) - \Phi^*\mathcal{A}(\alpha^*) = \begin{bmatrix}\sum_t n_t\alpha_t^{*\top} & \sum_t n_t\alpha^{*\top}_{t-1} & \cdots & \sum_t n_t\alpha^{*\top}_{t-p+1}\end{bmatrix}$ is the term generated by noise and should be bounded if noise terms are bounded.

$$||\Phi(\alpha) - \Phi^*||_{op} \leq ||\Phi(\alpha) - \Phi(\alpha^*)||_{op} + ||\Phi(\alpha^*) - \Phi^*||_{op} \tag{56}$$

$$\leq (\frac{2B_\alpha\sqrt{pT}}{\lambda_\mathcal{A}} + \frac{2B_\alpha^3\sqrt{p^3T^3}}{\lambda_\mathcal{A}^2})||\alpha - \alpha^*||_2 + ||\Phi(\alpha^*) - \Phi^*||_{op} \tag{57}$$

**Statistical consistency** We now have

$$||\Phi(\alpha) - \Phi^*||_{op} \leq (\frac{2B_\alpha\sqrt{pT}}{\lambda_\mathcal{A}} + \frac{2B_\alpha^3\sqrt{p^3T^3}}{\lambda_\mathcal{A}^2})||\alpha - \alpha^*||_2 + ||\Phi(\alpha^*) - \Phi^*||_{op} \tag{58}$$

$$||\alpha - \alpha^*||_2 \leq \frac{4\lambda p B_\Phi B_\alpha T}{\mu_\alpha}||\Phi - \Phi^*||_{op} + \frac{||\nabla_\alpha\mathcal{L}(\alpha^*, \Phi^*)^\top||_2}{\mu_\alpha} \tag{59}$$

Assume $||\nabla_\alpha\mathcal{L}(\alpha^*, \Phi^*)^\top||_2 \to 0$ and $||\Phi(\alpha^*) - \Phi^*||_{op} \to 0$. (Note: Do it with $||\Phi(\alpha) - \Phi(\alpha^*)||_{op}$) $L_\Phi L_\alpha := (\frac{2B_\alpha\sqrt{pT}}{\lambda_\mathcal{A}} + \frac{2B_\alpha^3\sqrt{p^3T^3}}{\lambda_\mathcal{A}^2})(\frac{4\lambda pB_\Phi B_\alpha T}{\mu_\alpha}) < 1$ will lead to the statistical consistency.

**Remarks** Identifiablity.

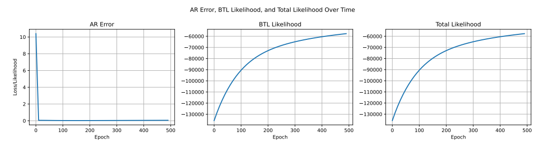
4. *Statistical sample complexity*: Finite time-step and no. of samples based guarantee on the estimation of true parameter.

# 7 Iterative Optimization Results

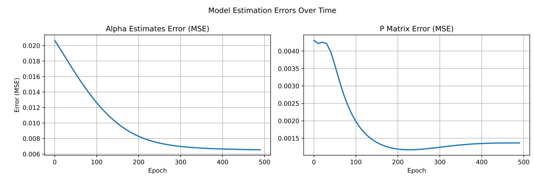
## 7.1 Parameters

| Parameter | Value |
|---|---|
| Erdos Renyi p | 1 |
| Number of players | 100 |
| AR order | 10 |
| Number of timesteps | 30 |
| AR $\sigma$ | 1e-1 |
| N_grad_descent | 10 |
| $\lambda$ | 10 |
| lr | 1e-3 |

(a) Experiment parameters. Fixed autoregressive process so that it doesn't decay rapidly.




(b) Likelihoods over time.




(c) True errors over time.

Figure 4: Iterative optimization results: (a) parameter settings, (b-c) likelihood and error evolution.

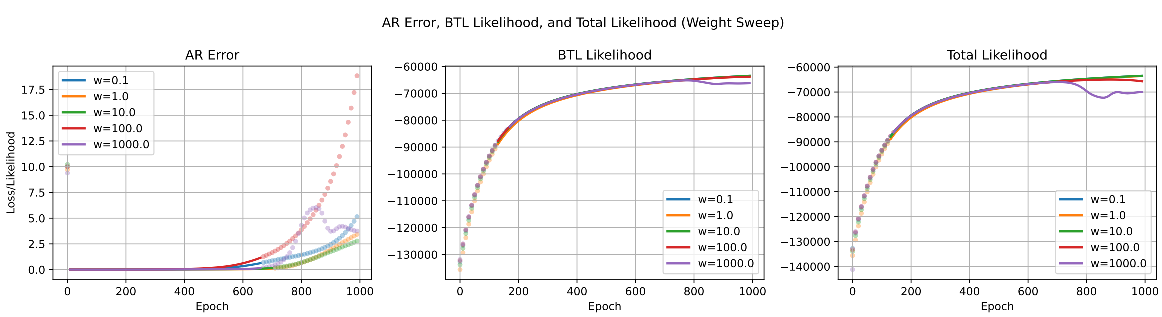



Figure 5: Sweep over various weights with first player fixed to 0 skill.

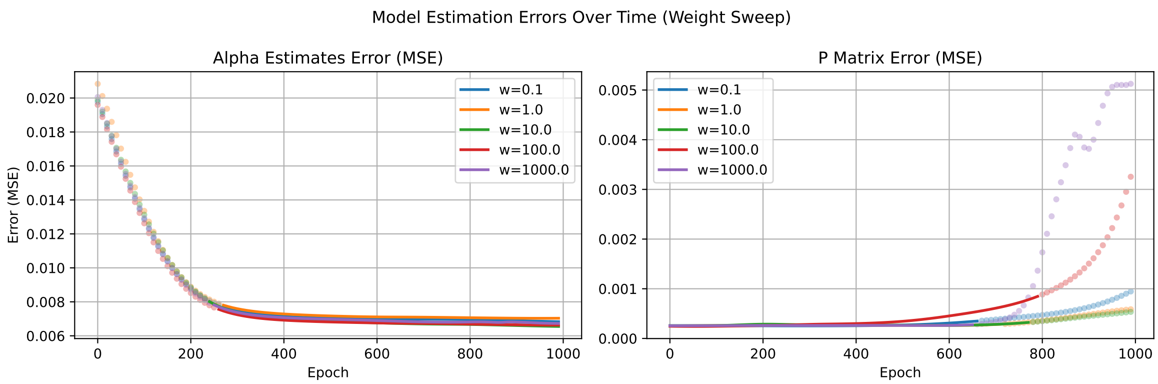



Figure 6: Errors with sweep over weights.

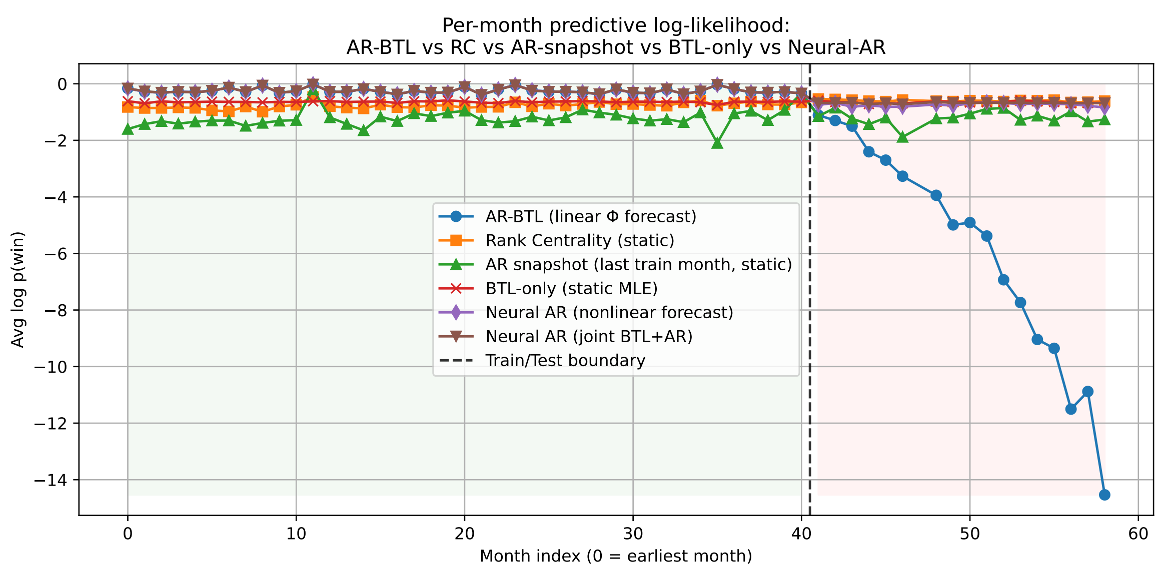



Figure 7: Tennis dataset